FX Research Lab — research protocol report

Strategy 1.1 (strategy_v1_1.yaml) · EURUSD · 2024-01-01 → 2024-12-31 · generated 2026-10-28 17:56 UTC · EURUSD_2024.csv

Educational research tool. This software is an educational research and backtesting tool. It is not investment advice, not a trading signal service, and it has no broker connection or live execution. All results are simulated on historical data using the recorded strategy and execution assumptions. Simulated results are not evidence of a profitable edge and historical performance does not guarantee future results. The author is not a licensed or registered financial adviser. Trading leveraged foreign exchange involves a high risk of loss; nothing here should be used to place real trades.

Reproducibility

Field	Value
Dataset SHA-256	65a7a807004594d4a14e4408ce4abc71db1bc8f2d6c73e72a19f5ca25acde62d
Configuration hash	307907477d561006cdb6ec34e39b019ff9245de679d8a9f410f958fef1098574
Execution model	first_executable_m1_price_after_bos
Spread sources	HISTORICAL_SPREAD, FIXED_CONFIGURED_SPREAD, NO_SPREAD_DATA
Slippage	0.1 pips
Timezone database	2026.5
Software	0.5.0 · source-sha256:29ff5ab67875481c0f492df189079cf3ca8bc6b0a52693c4da4e2aaad34bb945

Data

Rows	Gaps	Duplicates	Invalid	Weekend rows	Spread column	Source timezone
372774	1617	0	0	8005	yes	embedded_offsets

■ Data gaps detected; no candles were fabricated.
■ Weekend-dated bars detected; verify source sessions and market closures.

Research protocol

Fixed-rule baseline evaluated on independent periods, rolling out-of-sample folds, a bootstrap of the trade sequence, one-at-a-time parameter variations and random entries that pay the same costs. Only out-of-sample figures may be quoted, always next to the random-benchmark percentile and the sensitivity table.

Baseline

Period	Trades	Expectancy	Total	Win rate	PF	Max DD	Sharpe (ann.)
full sample	12	+0.20R	+2.35R	42 %	1.54	+3.01R	0.57

Independent chronological periods

Period	Trades	Expectancy	Total	Win rate	PF	Max DD	Sharpe (ann.)
train	8	+0.13R	+1.07R	38 %	1.33	+3.01R	0.40
validation	3	+0.44R	+1.31R	67 %	2.31	+1.00R	1.26
out_of_sample	1	-0.03R	-0.03R	0 %	0.00	+0.03R	-2.20

Out-of-sample metrics are the only ones that may be quoted as evidence; the baseline has no fitted parameters, so train/validation serve as independent replications.

Rolling out-of-sample (walk-forward)

9 folds · train 120 d, test 30 d. Concatenated OOS: 6 trades, +3.00R, expectancy +0.50R, 2/4 folds positive.

Period	Trades	Expectancy	Total	Win rate	PF	Max DD	Sharpe (ann.)
fold 1 OOS	0	+0.00R	+0.00R	0 %	—	+0.00R	—
fold 2 OOS	0	+0.00R	+0.00R	0 %	—	+0.00R	—
fold 3 OOS	2	+0.86R	+1.71R	50 %	7.01	+0.29R	2.92
fold 4 OOS	0	+0.00R	+0.00R	0 %	—	+0.00R	—
fold 5 OOS	2	+1.16R	+2.31R	100 %	—	+0.00R	3.81
fold 6 OOS	1	-1.00R	-1.00R	0 %	0.00	+1.00R	-3.46
fold 7 OOS	0	+0.00R	+0.00R	0 %	—	+0.00R	—
fold 8 OOS	0	+0.00R	+0.00R	0 %	—	+0.00R	—
fold 9 OOS	1	-0.03R	-0.03R	0 %	0.00	+0.03R	-7.94

Monte Carlo (trade-sequence bootstrap)

Paths	Trades / path	p(losing)	Final R p5	p50	p95	Max DD p50	Max DD p95
1000	12	28 %	-3.65R	+2.20R	+8.71R	+2.39R	+5.29R

Cost-matched random benchmark

Baseline expectancy +0.20R sits at percentile 76 of 200 simulations of random entries with identical stop distance, target, costs and session expiry (random median -0.07R/trade). Below the 95th percentile the result is indistinguishable from random timing.

Parameter sensitivity (one change at a time)

Parameter	Value	Trades	Expectancy	Total	Max DD
baseline	—	12	+0.20R	+2.35R	+3.01R
strategy:target.baseline	1.5R	12	+0.20R	+2.35R	+3.01R
strategy:target.baseline	2.0R	12	+0.20R	+2.35R	+3.01R
strategy:target.baseline	2.5R	12	+0.32R	+3.85R	+3.01R
strategy:target.baseline	3.0R	12	+0.34R	+4.06R	+3.01R
execution:slippage.baseline_pips	0.0	12	+0.21R	+2.46R	+3.00R

Parameter	Value	Trades	Expectancy	Total	Max DD
execution:slippage.baseline_pips	0.1	12	+0.20R	+2.35R	+3.01R
execution:slippage.baseline_pips	0.2	12	+0.19R	+2.26R	+3.00R
execution:slippage.baseline_pips	0.5	11	+0.19R	+2.06R	+3.00R
strategy:risk.baseline_equity_risk_fraction	0.0025	12	+0.20R	+2.35R	+3.01R
strategy:risk.baseline_equity_risk_fraction	0.005	12	+0.20R	+2.35R	+3.01R
strategy:risk.baseline_equity_risk_fraction	0.01	12	+0.20R	+2.35R	+3.01R
strategy:zones.expiry_m15_bars	4	12	+0.20R	+2.35R	+3.01R
strategy:zones.expiry_m15_bars	8	12	+0.20R	+2.35R	+3.01R
strategy:zones.expiry_m15_bars	12	12	+0.20R	+2.35R	+3.01R
strategy:setup.lifetime_m15_bars_from_sweep_confirmation	6	9	-0.07R	-0.65R	+2.01R
strategy:setup.lifetime_m15_bars_from_sweep_confirmation	8	12	+0.20R	+2.35R	+3.01R
strategy:setup.lifetime_m15_bars_from_sweep_confirmation	12	14	+0.36R	+4.98R	+3.00R
strategy:displacement.minimum_atr	0.75	12	+0.19R	+2.26R	+3.01R
strategy:displacement.minimum_atr	1.0	12	+0.20R	+2.35R	+3.01R
strategy:displacement.minimum_atr	1.25	11	+0.30R	+3.35R	+3.01R
strategy:stop.reference	sweep_extreme	13	+0.03R	+0.35R	+4.00R
strategy:stop.reference	zone_distal	12	+0.20R	+2.35R	+3.01R
strategy:stop.reference	leg1_origin	18	-0.21R	-3.74R	+6.49R
strategy:stop.max_distance_atr	3.0	12	+0.20R	+2.35R	+3.01R
strategy:stop.max_distance_atr	4.0	17	-0.09R	-1.55R	+5.00R
strategy:stop.max_distance_atr	5.0	19	-0.10R	-1.82R	+5.00R
strategy:liquidity.level_max_age_m15_bars	None	15	+0.22R	+3.29R	+4.01R
strategy:liquidity.level_max_age_m15_bars	960	10	+0.00R	+0.04R	+3.01R
strategy:liquidity.level_max_age_m15_bars	1920	12	+0.20R	+2.35R	+3.01R
strategy:liquidity.level_max_age_m15_bars	3840	12	+0.20R	+2.35R	+3.01R
multi:session	london_open_09:00-10:30	12	+0.20R	+2.35R	+3.01R
multi:session	new_york_open_15:00-16:30	8	-0.83R	-6.66R	+6.66R

One-at-a-time variations (and labelled multi-key session experiments); large swings between neighbouring values indicate parameter fragility. Experiments are not the baseline and must not be reported as such.

Limitations

- Simulated fills on historical M1 data; no live execution was involved.
- Costs follow config/execution.yaml; provider spreads are used when present.
- A positive result in one period is not evidence of a robust edge.
- Historical performance does not guarantee future results.

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